

CS 171: Problem Set 2

Due Date: September 17, 2026 at 8:59pm via Gradescope

1. Negligible/Non-negligible functions

Let $f, g : \mathbb{N} \rightarrow [0, 1]$ be negligible functions, let $p : \mathbb{N} \rightarrow \mathbb{R}$ be a polynomial such that $p(n) > 0$ for all $n \in \mathbb{N}$.

- (a) Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) + g(n)$. Prove that h is a negligible function.
- (b) Define $h : \mathbb{N} \rightarrow \mathbb{R}$ as $h(n) = f(n) \cdot p(n)$. Prove that h is a negligible function.

For each function below, either prove that it is negligible or prove that it is non-negligible (all logarithms are base 2).

- (c) $f(n) = n^{-100} + 2^{-n}$
- (d) $f(n) = 1.01^{-n}$
- (e) $f(n) = 2^{-(\log n)^2}$
- (f) $f(n) = e^{-\log^3 n} + e^{-\log^2 n} + e^{-\log n}$

2. 2-time security?

An encryption scheme ($\text{Gen}, \text{Enc}, \text{Dec}$) over message space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ is said to be 2-time perfectly secure if for any $(m_1, m_2) \in \mathcal{M} \times \mathcal{M}$ and $(m'_1, m'_2) \in \mathcal{M} \times \mathcal{M}$ such that $m_1 \neq m_2$ and $m'_1 \neq m'_2$ and for any $c_1, c_2 \in \mathcal{C}$ the following holds:

$$\Pr[\text{Enc}(K, m_1) = c_1 \wedge \text{Enc}(K, m_2) = c_2] = \Pr[\text{Enc}(K, m'_1) = c_1 \wedge \text{Enc}(K, m'_2) = c_2].$$

Note that in the above definition the key K is the same for encrypting m_1, m_2 (resp. m'_1, m'_2). Consider the following encryption scheme for the message space $\mathbb{Z}_{23}$.

- **Gen:** Sample two elements $a \xleftarrow{\$} \mathbb{Z}_{23}$ and $b \xleftarrow{\$} \mathbb{Z}_{23}$.
- **Enc** $((a, b), m)$: Output $c = a \cdot m + b \pmod{23}$.
- **Dec** $((a, b), c)$: Compute $m = (c - b) \cdot a^{-1} \pmod{23}$ if a is invertible. Otherwise, output error.

Show the following.

1. Prove that for any message $m \in \mathbb{Z}_{23}$,

$$\Pr[\text{Dec}(K, \text{Enc}(K, m)) = m] = \frac{22}{23}$$

2. Prove that this is 2-time secure.

3. Pseudorandom Functions

Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function (PRF). For the functions f' below, either prove that f' is a PRF (for all choices of f), or prove that f' is not a PRF.

(a) $f'_k(x) := f_k(0||x)||f_k(1||x)$.

(b) $f'_k(x) := f_k(0||x)||f_k(x||1)$.

4. Weak CPA Security

Consider a weaker definition of CPA security where in the indistinguishability experiment the adversary $\mathcal{A}$ is not given oracle access to $\text{Enc}_k(\cdot)$ after choosing m_0, m_1 . That is, $\mathcal{A}$ can only query $\text{Enc}_k(\cdot)$ in phase 1, but not in phase 2. We call this definition weak-CPA-security. Prove that weak-CPA-security is equivalent to CPA-security (i.e., Definition 3.22 in the textbook).

Hint: Begin by showing via a hybrid argument that any $\mathcal{A}$ interacting in the usual CPA game cannot distinguish whether its phase 2 queries are answered honestly (that is, if the response to the query m is $\text{Enc}_k(m)$ or an encryption of 0; $\text{Enc}_k(0)$ – something unrelated to m).